

Expectations and Moments

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Expectations and moments

The probability distribution defines weighted averages over the sample space, where the weight of each event is equal to its probability. These are called *expected values*.

Discrete case

For the discrete case,

$$\mathbb{E}[f(X)] = \sum_{k=1}^n f(x_k) p(x_k).$$

Continuous case

While for the continuous case,

$$\mathbb{E}[f] = \int_{-\infty}^{\infty} f(x) p(x) dx.$$

Mean of a distribution

The mean of the distribution is the expectation of the random variable itself. We denote

$$\mu \equiv \mathbb{E}[X] = \overline{X} = \langle X \rangle = \begin{cases} \sum x_k p(x_k), & \text{(discrete)} \\ \int_{-\infty}^{\infty} x p(x) dx, & \text{(continuous)} \end{cases}$$

In the case of an infinite sample space, whether continuous or discrete, the mean is not guaranteed to exist since the integral or the sum might not converge.

Moments of a distribution

The moments of a distribution are the expectation of powers of the random variable itself.

$$\mu_\ell \equiv \mathbb{E}[X^\ell] \equiv \langle X^\ell \rangle = \begin{cases} \sum x_k^\ell p(x_k), & (\text{discrete}) \\ \int_{-\infty}^{\infty} x^\ell p(x) dx, & (\text{continuous}) \end{cases}$$

If all the moments are known – and if they exist – they can be used to get the expectation of other functions using the **linearity** of the expectation operator:

$$\begin{aligned} \mathbb{E}[c f(X)] &= c \mathbb{E}[f(X)], \\ \mathbb{E}[f(X) + g(X)] &= \mathbb{E}[f(X)] + \mathbb{E}[g(X)] \end{aligned}$$

Variance and standard deviation

Of particular interest is the second moment, in combination with the mean, defining the **variance**

$$\begin{aligned} \sigma^2 = \text{Var}(X) &\equiv \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - \mu^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

Higher moments characterize properties of a distribution

Variance – dispersion measure based on the second moment:

$$\sigma^2 = \mathbb{E}[(X - \mu)^2] = \int (x - \mu)^2 p(x) dx$$

Skewness – asymmetry parameter based on third moments; dimensionless (normalized cumulant):

$$s \equiv \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3} = \mathbb{E}\left[\left(\frac{X - \mu}{\sigma}\right)^3\right]$$

Kurtosis – measure of tail "weights" in terms of fourth moments; zero for a Gaussian (excess kurtosis), bounded below by -1:

$$\kappa \equiv \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4} - 3$$

Covariance and correlation

For any two random variables, not necessarily independent or identically distributed, their covariance is defined as:

$$\text{Cov}(X, Y) \equiv \mathbb{E}[(X - \mu_x)(Y - \mu_y)] = \mathbb{E}[XY] - \mu_x \mu_y$$

The correlation is proportional to the covariance,

$$\rho(X, Y) = \text{Corr}(X, Y) \equiv \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \mathbb{E} \left[\left(\frac{X - \mu_x}{\sigma_x} \right) \left(\frac{Y - \mu_y}{\sigma_y} \right) \right]$$

Dividing the covariance by the standard deviations makes the correlation a pure number, and

$$-1 \leq \rho(X, Y) \leq +1$$

Common Distributions

Uniform Distribution

- **Standard form:**

$$p(x) = \begin{cases} 1, & x \in [0, 1] \\ 0, & \text{otherwise} \end{cases},$$

$$\text{Prob}(a < X < b) = \int_a^b p(x) dx = b - a,$$

$$F(x) = \int_{-\infty}^x p(x') dx' = x$$

- **Moments:**

$$\mu = \int_{-\infty}^{\infty} x p(x) dx = \int_0^1 x dx = \frac{1}{2},$$

$$\mu_\ell = \int_0^1 x^\ell dx = \frac{1}{\ell + 1},$$

$$\sigma^2 = \int_0^1 \left(x - \frac{1}{2} \right)^2 dx = \frac{1}{12}.$$

Binomial Distribution

- Identify events by total number of successes versus failures, independent of the order in which they occur.

- **Two parameters:** probability of success p , number of trials n .

$$f(k; n, p) = \binom{n}{k} p^k q^{n-k}, \quad \text{where}$$

$q = 1 - p$ is the probability of failure,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \binom{n}{n-k}.$$

- **Mean value (the hard way):**

$$\begin{aligned} \mu = \mathbb{E}[X] &= \sum_{k=0}^{\infty} k f(k; n, p) \\ &= \sum_{k=1}^n \frac{k n!}{k!(n-k)!} p^k q^{n-k} \\ &= np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{(n-1)-(k-1)} \\ &= np \sum_{k'=0}^{n'} f(k'; n', p), \quad \text{where } k' = k-1, \ n' = n-1 \\ &= np. \end{aligned}$$